

Fault Code: Multiple

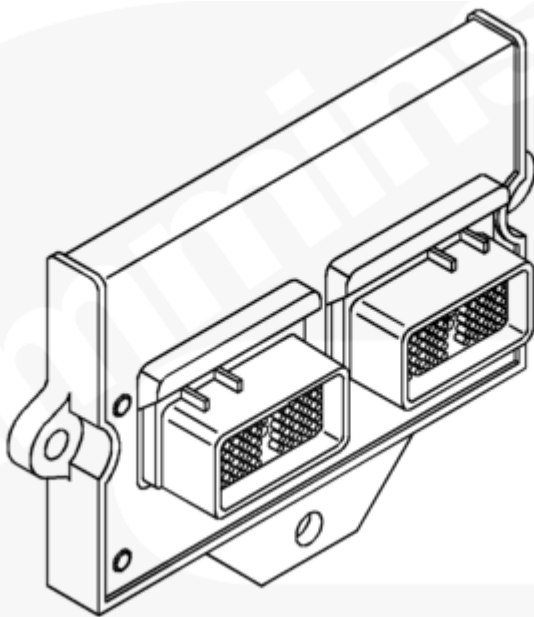
Multiple Fault Codes on the Engine Harness

 Printable Version

Overview

Codes	Reason	Effect
Fault Code: Multiple PID(P): SPN: FMI: Lamp: SRT:	Multiple fault codes generated due to a common supply or a return wire failure in the engine harness.	Multiple fault codes.

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Circuit Description

The electronic control module (ECM) supplies all of the engine pressure sensors on the engine harness with +5 VDC from pin 10. The ECM has common returns for all of the engine pressure sensors and temperature sensors on pin 19. A failure on either of these wires will cause multiple fault codes.

Component Location

Consult the engine diagrams.

Shoptalk

Look for an open circuit in the common supply and return wires and short circuits from battery or ground to the supply and return wires or defective ECM power supply.

A failed pressure sensor can cause multiple fault codes.

A failed pressure sensor can cause multiple active fault codes to go inactive once the engine has been started.

Warnings and Cautions

CAUTION

To reduce the possibility of damaging a new ECM, all other active fault codes must be investigated prior to replacing the ECM.

CAUTION

To reduce the possibility of pin and harness damage, use the following test lead when taking a measurement:

Part Number 3822758 - male Deutsch/AMP/Metri-Pack test lead.

Troubleshooting Summary

STEPS		SPECIFICATIONS
STEP 1.	Determine the set of multiple fault codes.	
	STEP 1A. Read the fault codes.	Fault Codes 122, 135, 144, 153, 221, and 719 are not active
	STEP 1B. Read the fault codes.	Fault Codes 122, 135, 221, 231, and 719 are not active
	STEP 1C. Read the fault codes.	Fault Codes 123, 141, 145, 154, 213, 222, 232, and 729 are not active
	STEP 1D. Read the fault codes.	Fault Codes 123, 141, 222, 232, and 729 are not active
STEP 2.	Check the engine harness.	

STEPS		SPECIFICATIONS
	STEP 2A. Inspect the ECM and engine harness connectors.	No damaged pins
	STEP 2B. Check for an open circuit.	Less than 10 ohms
	STEP 2C. Check for a short circuit from pin to pin.	More than 100k ohms
STEP 3.	Check the engine harness.	
	STEP 3A. Inspect the ECM and engine harness connectors.	No damaged pins
	STEP 3B. Check for an open circuit.	More than 100k ohms
STEP 4.	Check the engine harness.	
	STEP 4A. Inspect the ECM and engine harness connectors.	No damaged pins
	STEP 4B. Check for an open circuit.	Less than 10 ohms
	STEP 4C. Check for a short circuit to ground.	More than 100k ohms
	STEP 4D. Check for a short circuit from pin to pin.	More than 100k ohms
	STEP 4E. Check the battery supply voltage.	17.0 to 35.0 VDC
STEP 5.	Check the engine harness.	
	STEP 5A. Inspect the ECM and engine harness connectors.	No damaged pins
	STEP 5B. Check for an open circuit.	Less than 10 ohms
	STEP 5C. Check for a short circuit to ground.	More than 100k ohms
	STEP 5D. Check for a short circuit from pin to pin.	More than 100k ohms
	STEP 5E. Check the battery supply voltage.	17.0 to 35.0 VDC
STEP 6.	Clear the fault codes.	
	STEP 6A. Disable the fault code.	Multiple fault codes inactive

STEPS		SPECIFICATIONS
	STEP 6B. Clear the inactive fault codes.	All fault codes cleared

STEP 1. Determine the set of multiple fault codes.

STEP 1A. Read the fault codes.

Conditions :

- Turn keyswitch ON.

Action	Specification/Repair	Next Step
• Read the fault codes using INSITE™.	Fault Codes 122, 135, 144, 153, 221, and 719 are not active	1B
	Fault Code 122, 135, 144, 153, 221, or 719 is active	2A

STEP 1B. Read the fault codes.

Conditions :

- Turn keyswitch ON.

Action	Specification/Repair	Next Step
• Read the fault codes using INSITE™.	Fault Codes 122, 135, 221, and 719 not active	1C
	Fault Code 122, 135, 221, or 719 is active	3A

STEP 1C. Read the fault codes.

Conditions :

- Turn keyswitch ON.

Action	Specification/Repair	Next Step
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Read the fault codes using INSITE™.	Fault Codes 123, 141, 145, 154, 222, 232, and 729 not active	1D
	Fault Code 123, 141, 145, 154, 222, 232, or 729 is active	4A

STEP 1D. Read the fault codes.

Conditions : Turn keyswitch ON.		
Action	Specification/Repair	Next Step
Read the fault codes using INSITE™.	Fault Codes 123, 141, 222, 232, and 729 not active	2A
	Fault Code 123, 141, 222, 232, or 729 is active	5A

STEP 2. Check the engine harness.

STEP 2A. Inspect the ECM and engine harness connectors.

Conditions : - Turn keyswitch OFF. - Disconnect the engine harness from the ECM.		
Action	Specification/Repair	Next Step

• Corroded pins • Bent or broken pins • Pushed back or expanded pins • Moisture in or on the connector • Missing or damaged connector seals • Dirt or debris in or on the connector pins.	No damaged pins	2B
	Repair the damaged pins Flush the dirt, debris, or moisture from the connector pins. Repair or replace the engine harness, or replace the ECM, whichever has the damaged pins. • Flush the dirt, debris, or moisture from the connector pins using electrical contact cleaner, Part Number 3824510. • Install the appropriate connector seal if it is damaged or missing. • Repair the engine harness. Refer to Procedure 019-204 (/qs3/pubsys2/xml/en/procedures/99/99-019-204.html). • Replace the engine harness. Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/87/87-019-043.html). • Replace the ECM. Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/87/87-019-031.html). • Dry the connector by using an electrical contact cleaner, Part Number 3824510.	6A

STEP 2B. Check for an open circuit.

Conditions :

- Turn keyswitch OFF.
- Disconnect the engine harness from the ECM.
- Disconnect the engine harness in the multiple fault code circuit from the sensor.

Action	Specification/Repair	Next Step
• Measure the resistance from pin 19 in the engine harness connector to pin B on the engine wiring harness side of all sensors with active fault codes.	Less than 10 ohms	2C
	Repair or replace the engine harness • Repair the engine harness. Refer to Procedure 019-202 (/qs3/pubsys2/xml/en/procedures/99/99-019-202.html) or 019-204 (/qs3/pubsys2/xml/en/procedures/99/99-019-204.html). • Replace the engine harness. Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/87/87-019-043.html).	6A

STEP 2C. Check for a short circuit from pin to pin.**Conditions :**

- Turn keyswitch OFF.
- Disconnect the engine harness from the ECM.
- Disconnect the OEM harness from the ECM.
- Disconnect the engine harness in the multiple fault code circuit from all the sensors.

Action	Specification/Repair	Next Step

Measure the resistance from pin 19 of the engine harness connector to all other pins in the connector, and to all pins in the OEM harness connector.	More than 100k ohms	6A
	Repair or replace the engine harness - Repair the engine harness. Refer to Procedure 019-204 (/qs3/pubsys2/xml/en/procedures/99/99-019-204.html). - Replace the engine harness. Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/87/87-019-043.html).	6A

STEP 3. Check the engine harness.

STEP 3A. Inspect the ECM and engine harness connectors.

Conditions :

- Turn keyswitch OFF.
- Disconnect the engine harness from the ECM.

Action	Specification/Repair	Next Step

• Corroded pins • Bent or broken pins • Pushed back or expanded pins • Moisture in or on the connector • Missing or damaged connector seals • Dirt or debris in or on the connector pins.	No damaged pins	3B
	Repair the damaged pins Flush the dirt, debris, or moisture from the connector pins. Repair or replace the engine harness, or replace the ECM, whichever has the damaged pins. • Flush the dirt, debris, or moisture from the connector pins using electrical contact cleaner, Part Number 3824510. • Install the appropriate connector seal if it is damaged or missing. • 019-204 (/qs3/pubsys2/xml/en/procedures/99/99-019-204.html). • Replace the engine harness. Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/87/87-019-043.html). • Replace the ECM. Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/87/87-019-031.html).	6A

STEP 3B. Check for an open circuit.

Conditions :

- Turn keyswitch OFF.
- Disconnect the engine harness from the ECM.
- Disconnect the engine harness in the multiple fault code circuit from all the sensors.

Action	Specification/Repair	Next Step
Measure the resistance from pin 19 of the engine harness connector to pin B of the harness side of all sensors with active fault codes.	More than 100k ohms	6A
	Repair or replace the engine harness - Repair the engine harness. Refer to Procedure 019-204 (/qs3/pubsys2/xml/en/procedures/99/99-019-204.html). - Replace the engine harness. Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/87/87-019-043.html).	6A

STEP 4. Check the engine harness.

STEP 4A. Inspect the ECM and engine harness connectors.

Conditions :

- Turn keyswitch OFF.
- Disconnect the engine harness from the ECM.

Action	Specification/Repair	Next Step
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	No damaged pins	4B
	Repair the damaged pins Flush the dirt, debris, or moisture from the connector pins. Repair or replace the engine harness, or replace the ECM, whichever has the damaged pins. • Flush the dirt, debris, or moisture from the connector pins using electrical contact cleaner, Part Number 3824510. • Install the appropriate connector seal if it is damaged or missing. • Repair the engine harness. Refer to Procedure 019-204 (/qs3/pubsys2/xml/en/procedures/99/99-019-204.html). • Replace the engine harness. Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/87/87-019-043.html). • Replace the ECM. Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/87/87-019-031.html).	6A
• Corroded pins • Bent or broken pins • Pushed back or expanded pins • Moisture in or on the connector • Missing or damaged connector seals • Dirt or debris in or on the connector pins.		

STEP 4B. Check for an open circuit.

Conditions :

- Turn keyswitch OFF.
- Disconnect the engine harness from the ECM.
- Disconnect the engine harness in the multiple fault code circuit from all the pressure sensors.

Action	Specification/Repair	Next Step
• Measure the resistance from pin 10 in the engine harness connector to pin A on the engine wiring harness side of a pressure sensor connector.	Less than 10 ohms	4C
	Repair or replace the engine harness • Repair the engine harness. Refer to Procedure 019-204 (/qs3/pubsys2/xml/en/procedures/99/99-019-204.html). • Replace the engine harness. Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/87/87-019-043.html).	6A

STEP 4C. Check for a short circuit to ground.**Conditions :**

- Turn keyswitch OFF.
- Disconnect the engine harness from the ECM.
- Disconnect the engine harness in the multiple fault code circuit from all the sensors.

Action	Specification/Repair	Next Step

• Measure the resistance from pin 10 of the engine harness connector to the engine block ground. • Measure the resistance from pin 23 of the engine harness connector to the engine block ground. • Measure the resistance from pin 34 of the engine harness connector to the engine block ground.	More than 100k ohms	4D
	Repair or replace the engine harness • Repair the engine harness. Refer to Procedure 019-204 (/qs3/pubsys2/xml/en/procedures/99/99-019-204.html). • Replace the engine harness. Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/87/87-019-043.html).	6A

STEP 4D. Check for a short circuit from pin to pin.

Conditions :

- Turn keyswitch OFF.
- Disconnect the engine harness from the ECM.
- Disconnect the OEM harness from the ECM.
- Disconnect the engine harness in the multiple fault code circuit from all the sensors.

Action	Specification/Repair	Next Step
• Measure the resistance from pin 10 of the engine harness connector to all other pins in the connector, and to all pins in the OEM harness connector.	More than 100k ohms	4E
	Repair or replace the engine harness • Repair the engine harness. Refer to Procedure 019-204 (/qs3/pubsys2/xml/en/procedures/99/99-019-204.html). • Replace the engine harness. Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/87/87-019-043.html).	6A

STEP 4E. Check the battery supply voltage.

Conditions :

- Turn keyswitch OFF.
- Disconnect the engine harness from the ECM.
- Disconnect the engine harness in the multiple fault code circuit from all the sensors.

Action	Specification/Repair	Next Step
• Measure the voltage from pin 50 of the engine harness connector to engine block ground.	17.0 to 35.0 VDC	5A
	Correct battery supply voltage problem.	6A

STEP 5. Check the engine harness.

STEP 5A. Inspect the ECM and engine harness connectors.

Conditions :

- Turn keyswitch OFF.
- Disconnect the engine harness from the ECM.

Action	Specification/Repair	Next Step
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	No damaged pins	5B
	Repair the damaged pins Flush the dirt, debris, or moisture from the connector pins. Repair or replace the engine harness, or replace the ECM, whichever has the damaged pins. • Flush the dirt, debris, or moisture from the connector pins using electrical contact cleaner, Part Number 3824510. • Install the appropriate connector seal if it is damaged or missing. • Repair the engine harness. Refer to Procedure 019-204 (/qs3/pubsys2/xml/en/procedures/99/99-019-204.html). • Replace the engine harness. Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/87/87-019-043.html). • Replace the ECM. Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/87/87-019-031.html).	6A
• Corroded pins • Bent or broken pins • Pushed back or expanded pins • Moisture in or on the connector • Missing or damaged connector seals • Dirt or debris in or on the connector pins.		

STEP 5B. Check for an open circuit.

Conditions :

- Turn keyswitch OFF.
- Disconnect the engine harness from the ECM.
- Disconnect the engine harness in the multiple fault code circuit from all the pressure sensors.

Action	Specification/Repair	Next Step
• Measure the resistance from pin 10 in the engine harness connector to pin A on the engine wiring harness side of all pressure sensors with active fault codes.	Less than 10 ohms	5C
	Repair or replace the engine harness • Repair the engine harness. Refer to Procedure 019-204 (/qs3/pubsys2/xml/en/procedures/99/99-019-204.html). • Replace the engine harness. Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/87/87-019-043.html).	6A

STEP 5C. Check for a short circuit to ground.**Conditions :**

- Turn keyswitch OFF.
- Disconnect the engine harness from the ECM.
- Disconnect the engine harness in the multiple fault code circuit from all the sensors.

Action	Specification/Repair	Next Step

• Measure the resistance from pin 10 of the engine harness connector to the engine block ground. • Measure the resistance from pin 23 of the engine harness connector to the engine block ground. • Measure the resistance from pin 34 of the engine harness connector to the engine block ground.	More than 100k ohms	5D
	Repair or replace the engine harness • Repair the engine harness. Refer to Procedure 019-204 (/qs3/pubsys2/xml/en/procedures/99/99-019-204.html). • Replace the engine harness. Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/87/87-019-043.html).	6A

STEP 5D. Check for a short circuit from pin to pin.

Conditions :

- Turn keyswitch OFF.
- Disconnect the engine harness from the ECM.
- Disconnect the OEM harness from the ECM.
- Disconnect the engine harness in the multiple fault code circuit from all the sensors.

Action	Specification/Repair	Next Step
• Measure the resistance from pin 10 of the engine harness connector to all other pins in the connector, and to all pins in the OEM harness connector.	More than 100k ohms	5E
	Repair or replace the engine harness • Repair the engine harness. Refer to Procedure 019-204 (/qs3/pubsys2/xml/en/procedures/99/99-019-204.html). • Replace the engine harness. Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/87/87-019-043.html).	6A

STEP 5E. Check the battery supply voltage.

Conditions :

- Turn keyswitch OFF.
- Disconnect the engine harness from the ECM.
- Disconnect the engine harness in the multiple fault code circuit from all the sensors.

Action	Specification/Repair	Next Step
• Measure the voltage from pin 50 of the engine harness connector to engine block ground.	17.0 to 35.0 VDC	6A
	Correct battery supply voltage problem.	6A

STEP 6. Clear the fault codes.

STEP 6A. Disable the fault code.

Conditions :

- Connect all components.
- Turn keyswitch ON.

Action	Specification/Repair	Next Step
• Start the engine and let it idle for 1 minute. • Verify that multiple fault codes are inactive.	Multiple fault codes inactive	6B
	Return to the troubleshooting steps or contact your local Cummins Authorized Repair Location if all the steps have been completed and checked again.	1A

STEP 6B. Clear the inactive fault codes.

Conditions :

- Connect all components.
- Turn keyswitch ON.

Action	Specification/Repair	Next Step
Erase the inactive fault codes using INSITE™.	All fault codes cleared	Repair complete
	Troubleshoot any remaining active fault codes.	Appropriate troubleshooting charts

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